

Muting (Don) Ma, Ph.D.

Operations Researcher • Optimization and Statistical Learning

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Summary

Operations researcher with a Ph.D. in transportation engineering and eight years of work in deterministic and stochastic modeling for connected, autonomous, and electric vehicle systems. Five peer-reviewed papers in *ITOR*, *Pattern Recognition*, *Transportation Research Part C*, NCHRP IDEA, and *TRIP*, including a dynamic program that cuts scheduling time 91% versus a state-of-the-art DP and 99% versus Gurobi. Lead researcher on \$509,059 in competitive funding; open implementations in Python, R, and C#; mentor to undergraduate, master's, and doctoral students.

Education

University of Louisville

Louisville, KY

Ph.D., Transportation Engineering

2022

Dissertation: Optimal scheduling of connected and autonomous vehicles at reservation-based intersections. Awards: Grosscurth Fellowship (2017); Doctoral Dissertation Completion Award (2022).

Additional graduate coursework at the University of Alabama in linear and integer programming (OM 600, OM 603), stochastic decision models (OM 601), mathematical statistics (ST 554), predictive modeling (BAN 501), inventory (OM 523), and game theory (EC 660). Foundations of Optimization I (IE 610) and Data Mining (CECS 632) at Louisville.

Peer-Reviewed Publications

- Li, Z., **Ma, M.***, Yavuz, M. Reservation-based intersection scheduling using dynamic programming with dominance pruning. *International Transactions in Operational Research*, 2026. [DOI](#) • [Code](#). DP² with analytical lower-bound node elimination; $O(n^2)$ worst case and $\Omega(n)$ best case; 91%/99% speedup versus prior DP and Gurobi.
- **Ma, M.***, Yavuz, M., Hudnall, M., Wang, Q. C2SLM: a correlation-based clustering-assisted sparse learning model for electric vehicle market demand forecasting. *Pattern Recognition*, 170, 111984, 2025. [DOI](#) • [Code](#). Correlation-polarized features, panel clustering, and caLasso; 20.1% lower test MeAE versus glmmLasso.
- **Ma, M.**, Li, Z. A speed-maximization trajectory optimization model on a reservation-based intersection control system. *Transportation Research Part C: Emerging Technologies*, 154, 104266, 2023. [DOI](#). Discrete-time MIP with queue-theoretic objective derivation and piecewise-feasible speed profiles.
- Li, Z., **Ma, M.** A real-time proactive intersection safety monitoring and visualization system based on radar sensor data. *NCHRP IDEA Project 217*, 2022. [TRB](#) • [Report](#). 24 GHz Doppler fusion with TTC conflict detection: 4.8% severity error, 80% true-positive rate, 100% true-negative rate.
- **Ma, M.**, Li, Z. A time-independent trajectory optimization approach for connected and autonomous vehicles under reservation-based intersection control. *Transportation Research Interdisciplinary Perspectives*, 9, 100312, 2021. [DOI](#). Integer linear program with variable acceleration, cell-based reservation, and BATCH/ZONE high-demand control.

* Corresponding author.

Academic Appointments

University of Alabama, Alabama Center for the Advancement of AI

Tuscaloosa, AL

Graduate Assistant

2026 - Present

- Writing NSF Operations Engineering and NHTSA Innovative Traffic Safety Enforcement proposals on knowledge-grounded stochastic optimization, traffic safety analytics, and program evaluation.

University of Alabama, Culverhouse College of Business

Tuscaloosa, AL

Graduate Teaching Assistant

2025 - 2026

- Assisted in teaching OM 310 Introduction to Management Science (40 undergraduates), OM 524 Manufacturing Scheduling and Control (9 graduates), and OM 525 Effective Quality Management (26 graduates).
- Teaching demos: [weak-supervision DistilBERT classification](#) and [CoopetAI game-theoretic market dashboards](#).

University of Alabama, Institute of Data and Analytics

Tuscaloosa, AL

Postdoctoral Research Associate

2022 - 2025

- Developed C2SLM for panel EV demand and system-dynamics models of technology adoption; lead researcher on the Alabama Mobility and Power Center (\$287,751).
- Led an interdisciplinary team of 8+ and mentored undergraduate, master's, and doctoral researchers on sparse learning, data analysis, and academic writing.

University of Louisville, Department of Civil and Environmental Engineering

Louisville, KY

Graduate Research Assistant

2017 - 2022

- Developed DP² scheduling, speed-maximization MIP trajectories, and the IPSV radar safety monitor under NSF I/UCRC, NCHRP IDEA 217, and KYTC HSIP.

Research Grants

- Alabama Mobility and Power Center, \$287,751 (2022 - 2025). Lead researcher. EV market analysis and demand forecasting.
- NCHRP IDEA Project 217, \$137,000 (2019 - 2022). Graduate research assistant. Real-time intersection safety monitoring.
- KYTC Highway Safety Improvement Program, \$39,318 (2019 - 2021). Graduate research assistant. Safety countermeasures.
- NSF Industry-University Cooperative Research Centers, \$45,990 (2017 - 2022). Graduate research assistant. Connected-vehicle intersection control.
- Total competitive funding: \$509,059.

Open-Source Software

DP² ([GitHub](#)) • **C2SLM** ([GitHub](#)) • **SLM** ([GitHub](#)) • **BiCAVO** ([Zenodo](#)) • **LP Octahedron** ([GitHub](#))

Python and R companions to the ITOR, Pattern Recognition, and trajectory papers, plus a two-stage stochastic MIP with Benders cuts and Gurobi notebooks on primal-dual LP.

Awards and Service

Awards: 2nd Place, Culverhouse Data Analytics Summit Poster Competition (2025); Doctoral Dissertation Completion Award (2022); Grosscurth Fellowship (2017).

Reviewer: *International Transactions in Operational Research*; *IEEE Transactions on Intelligent Transportation Systems*; *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; *Scientific Data*; *Soft Computing*; *PLOS One*; *International Journal of Energy Research*; *Energy Engineering*; *Energy Systems*.

Memberships: INFORMS; Transportation Research Board.

Methods

Linear and integer programming • Dynamic programming • Dominance pruning • Benders decomposition • Stochastic programming • Sparse learning • Panel econometrics • Game theory • Simulation-optimization • Python, R, C#, C++, Java, SQL • Gurobi, CPLEX